

# Measurement in the Age of AI: Fundamentals and Uncertainty Quantification

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UniMORE AI Center – Via Pietro Vivarelli 2, Modena

## ABSTRACT:

Like any science and engineering field, Instrumentation and Measurement (I&M) is currently experiencing the impact of the recent rise of Artificial Intelligence (AI) as an applied tool. In fact, there is an intertwined relationship between the two: measurement is used to collect data, which are then used to train AI models, which in turn are used for indirect measurement. The applications are vast: medical diagnosis, surveillance, fault detection, condition monitoring, etc. At the same time, uncertainty, which is a fundamental component of measurement, must be quantified for risk management and better decision making in AI-assisted measurement systems. In this talk, we will learn how AI is used for indirect measurement and how to quantify the uncertainty of AI-assisted measurements to design more reliable and trustworthy measurement systems. We will cover AI regression, classification, and Large Language Models (LLM). Additionally, we will provide a unified taxonomy of uncertainty quantification methods and point to some measurement-standard noncompliance in AI literature, which readers need to be aware of. Finally, we go over a few specific examples from existing literature.



## BIOGRAPHY OF THE SPEAKER:

Shervin Shirmohammadi received his Ph.D. in Electrical Engineering in 2000 from the University of Ottawa, Canada, and after spending 3 years in the industry as a senior architect and project manager, joined as Assistant Professor the same University, where since 2012 he has been a Full Professor with the School of Electrical Engineering and Computer Science. He is Director of the Discover Laboratory, doing research in AI-assisted measurements and sensing systems, specifically fundamentals of AI in measurement, human behavior and activity sensing, multimedia systems and network measurements, IoT measurements, and biomedical measurements. The results of his research, funded by more than \$28 million from public and private sectors, have led to over 450 publications, over 80 researchers trained at the postdoctoral, PhD, and Master's levels, 40 patents and technology transfers to the private sector, and five Best Paper awards. He was the Founding Editor-in-Chief of the IEEE Open Journal of Instrumentation and Measurement in 2022 and 2023, the Editor-in-Chief of the IEEE Transactions on Instrumentation and Measurement from 2017 to 2021, and the Associate Editor-in-Chief of IEEE I&M Magazine in 2014 and 2015, and is currently on the latter's editorial board. He has been on the Administrative Committee of the IEEE Instrumentation and Measurement Society (IMS) since 2014, currently serves as IMS's President, and was a member of the IEEE I2MTC Board of Directors from 2014 to 2016. Dr. Shirmohammadi is an IEEE Fellow "for contributions to multimedia systems and network measurements", and recipient of the 2019 George S. Glinski Award for Excellence in Research, the 2021 IEEE IMS Distinguished Service Award, and the 2023 IEEE IMS Technical Award "for contributions to the advancement of machine learning-assisted measurements".